

Archibald Qian

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Summary

- Data scientist with an MSc in Statistics (GPA 3.9/4.0) and hands-on experience in electricity load forecasting, power-system analysis, and energy market data.
- Built ARIMA/regression demand-forecasting models (<12% MAPE) with forecast-accuracy and deviation analysis at BBA Engineer, plus automated Python/PSS®E transmission analysis workflows.
- Strong Python and SQL background with hands-on experience using LLMs, Sentence Transformers, NER, PyTorch, AWS SageMaker, Databricks, and MLflow to process large datasets and deploy analytical solutions.
- Exposure to Ontario electricity-sector data through an independent IESO market-data simulator project.

WORK EXPERIENCE

BBA Engineer Inc.

Calgary, AB – Toronto, ON

Power Systems & Industrial Data Analytics

Sept. 2024 – Present

- Built feeder-level electricity load forecasting models for Oakville Hydro using ARIMA and regression with temperature, humidity, calendar, and COVID-period features, achieving <12% MAPE; aggregated underperforming feeders into larger zones with variance modeling, and performed forecast-accuracy and deviation analysis.
- Applied quantile regression to forecast 90th-percentile peak load, complementing point forecasts with probabilistic peak estimates.
- Developed an **hourly battery-dispatch optimization model** formulated as a linear program in Python (PuLP) for IESO, co-optimizing day-ahead/real-time energy arbitrage with spinning reserve provision against 20+ years of client-provided forecasted price data; built Power BI dashboards visualizing annual revenue and hourly dispatch.
- Automated **N-0/N-1 transmission** injection-capacity analysis: built Python/PSS®E workflows on MMWG base cases with LT1/LT2 (window 1 & 2) topology updates and Hydro One transmission-line GIS data; used bisection search adding 0–300 MW generation per circuit for N-0 maximum capacity and PSS®E ACCC for N-1 contingencies; collaborated with the GIS team to build an ArcGIS results-visualization tool.
- Built statistical models predicting risk and failure probability for 100,000+ power transmission assets from operational and maintenance data (automated ETL from AccessDB into the modeling backend), guiding \$1M+ in maintenance planning.
- Researched and developed **EV charging load profiles** across Canadian regions, demonstrating ~20% reduction in peak-demand impact from managed charging strategies.
- Researched EV-driven load growth across Quebec, Ontario, and BC for a CEATI project; analyzed high-penetration trends under the 2035 gas-vehicle ban and quantified how time-of-use (TOU) rates moderate load increment.
- Supported cybersecurity and infrastructure initiatives in regulated industrial environments, including SCADA/OT asset inventories, risk assessments, and network segmentation.

Intact Financial Corporation

Calgary, AB

Data Scientist Intern

Oct. 2023 – Jan. 2024

- Developed and evaluated NLP classification models using Large Language Models and Sentence Transformers (PyTorch) to classify customer call transcripts into 10+ operational categories, achieving >60% F1 score and enabling automated analysis of customer interactions.
- Built end-to-end **ML pipelines in AWS SageMaker**, integrating large-scale transcript data from Amazon S3 and tracking model performance and experiments with MLflow.

- Developed scalable ETL pipelines in Python and SQL to transform and process large-scale call transcript datasets for model training, inference, and downstream customer-service analytics.
- Implemented rule-based Named Entity Recognition (NER) pipelines in Python to extract structured entities from unstructured call transcripts, enabling downstream analytics and feature generation.

Cenozon Inc.

Calgary, AB

Data Scientist Intern

July 2023 – Oct. 2023

- Analyzed oil pipeline operational datasets including production rates, fluid composition, and pipeline network metrics using SQL and Python.
- Developed machine learning regression models (Linear Regression, Decision Tree, Neural Networks) to predict pipeline corrosion rates and infrastructure risk, achieving >90% accuracy relative to engineering baseline calculations.
- Built data processing pipelines to integrate Excel-based inspection readings with production databases, improving data completeness and enabling automated analysis workflows.

PROJECTS

Grid Market Simulator (Meta Grid) | Personal Project

Dec. 2025 - Present

- Built automated ETL pipelines ingesting IESO market data (demand, prices) into a structured store with an interactive dashboard for exploratory market analysis;
- Designed short-term load-forecasting and battery-dispatch optimization layers (in development).
- Published a technical EDA of 20+ years of Ontario hourly load and weather data (~214k observations, 2002–2026), documenting nonlinear temperature-load relationships, humidity and calendar effects, and evolving heat sensitivity.

RESEARCH EXPERIENCE

Machine Learning Researcher | Cross-validation Correction

Sept. 2021 – Sept. 2023

- Developed a statistical structure using covariance between the Training set and the Test set to adjust the Cross-Validation Error.
- Explained and tested corrected CV error for the models of Linear Mixed-Effects Model (LMM), Best Linear Unbiased Predictor (BLUP), Ridge regression, and Bayesian Selection Linear Mixed Model (BSLMM).
- Published: The Bias of Using Cross-Validation in Genomic Predictions and Its Correction

EDUCATION

MSc in Statistics (GPA: 3.9/4) University of Calgary

Calgary, Canada

Thesis: Cross-Validation Correction of Machine Learning Algorithms in Genomic Data

Sept. 2021 – Sept. 2023

BSc in Applied Mathematics (GPA: 3.7/4)

Shenzhen, China

Southern University of Science and Technology

SKILLS

AI / Machine Learning: LLMs, Retrieval-Augmented Generation (RAG), ReAct Agents, Tool Calling, Vector Databases (Weaviate), Sentence Transformers, NLP, Named Entity Recognition, Statistical Modeling, Regression, Time Series, Experimental Design / A/B Testing, Model Evaluation, Scikit-learn, PyTorch, XGBoost

Programming & Data: Python, SQL, R, NumPy, Pandas, SQL Server, PostgreSQL, ETL, Power BI

Cloud & MLOps: GCP (Cloud Run, Cloud SQL), AWS SageMaker, Amazon S3, Azure Databricks, MLflow, Git, Docker, Machine Learning Model Development, Machine Learning Deployment, Machine Learning Pipelines